Suez Canal
Branch

etisalat and
In collaboration with

RANPilotAI

AI-Powered RAN Performance Optimization Platform

Graduation Project ▪ 2026

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Intake 46 ▪ Telco-Cloud Track ▪ Information Technology Institute

Agenda

Tasneem Amein

Problem Statement ▪ Proposed Solution ▪ Overall Architecture ▪ KPI Degradation Analyzer

Problem Statement

[Fill in your content here]

Project Motivation & Objectives

[Fill in your content here]

Proposed Solution Overview

[Fill in your content here]

Overall System Architecture

Screenshot not yet added

`screenshots/tasneem/system_architecture.png`

Telecom Hub — Landing Page

Screenshot not yet added

`screenshots/tasneem/hub_landing.png`

KPI Degradation Analyzer — Module Overview

[Fill in your content here]

KPI Categories Monitored

[Fill in your content here]

Detection Pipeline & Root Cause Analysis

[Fill in your content here]

KPI Degradation Analyzer — Interface

Screenshot not yet added

`screenshots/tasneem/kpi_degradation_ui.png`

KPI Degradation — Sample Results

Screenshot not yet added

`screenshots/tasneem/kpi_degradation_results.png`

Ibrahim Samy

KPI Forecasting Engine ▪ Models ▪ Feature Engineering ▪ Accuracy & Results

Forecasting Module Overview & Motivation

[Fill in your content here]

Forecasting Approach — XGBoost & Holt-Winters

[Fill in your content here]

Feature Engineering & Seasonality Decomposition

[Fill in your content here]

Forecasting Module — Interface

Screenshot not yet added

`screenshots/ibrahim/forecasting_ui.png`

Model Evaluation & Accuracy Metrics

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`screenshots/ibrahim/accuracy_metrics.png`

7-Day KPI Forecast — Output

Screenshot not yet added

`screenshots/ibrahim/forecast_chart.png`

XGBoost Feature Importance

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Khaled Mogahed

Optimization Action Analyzer ▪ Configuration Tracking ▪ Impact Quantification

Action Analyzer — Module Overview

[Fill in your content here]

Configuration Tracking with DoIt

[Fill in your content here]

Action Analyzer — Interface

Screenshot not yet added

`screenshots/khaled/action_analyzer_ui.png`

Action Timeline View

Screenshot not yet added

`screenshots/khaled/action_timeline.png`

Before / After KPI Impact Analysis

Before Configuration Change

Screenshot not yet added

`screenshots/khaled/before_change.png`

After Configuration Change

Screenshot not yet added

`screenshots/khaled/after_change.png`

KPI Delta Bar Chart

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Mahmoud Ashraf

Telecom RAG Assistant ▪ Pipeline ▪ Interface ▪ Anti-Hallucination Strategy

RAG Assistant — Overview & Motivation

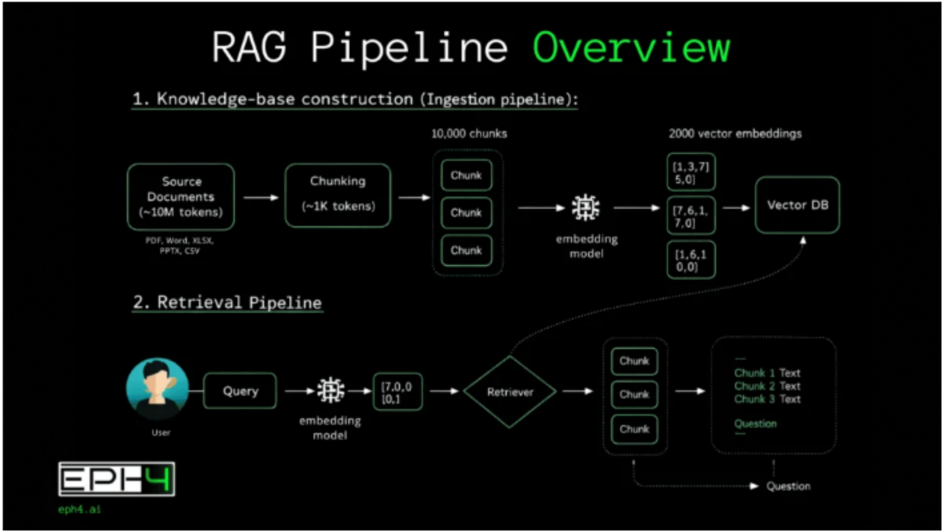
The Problem

- Engineers waste hours manually searching 3GPP PDF specs
- Keyword search fails when wording differs from the spec
- General LLMs **hallucinate** protocol details
- No tool provides citation-backed, spec-grounded answers

Our Solution

- **Engineering-grade RAG** purpose-built for 3GPP
- Fully **local**: embeddings, vector store, re-ranking
- Every answer cites: TS number, Release, Section, page
- Integrated into RANPilotAI — no tool switching

RAG Pipeline Architecture



Pipeline — Key Engineering Decisions

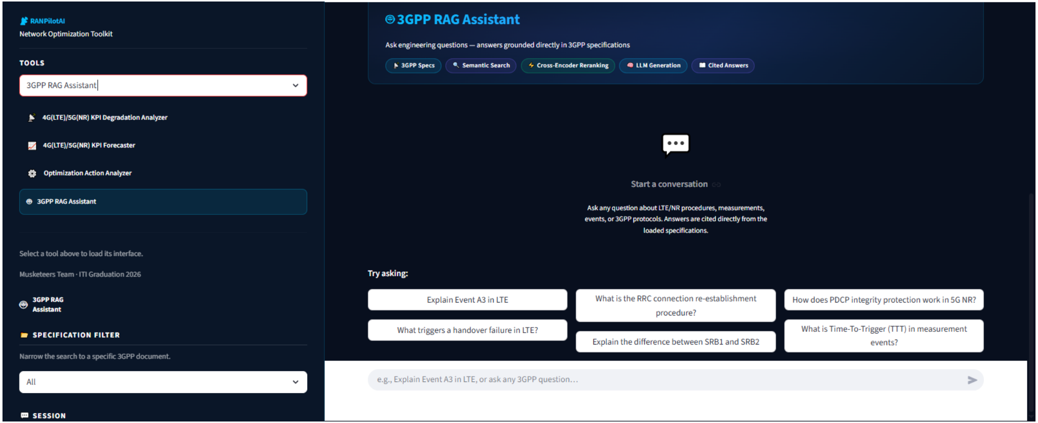
Ingestion & Chunking

- **Section-aware** PDF parsing (PyMuPDF) splits on 3GPP section boundaries — formulas and ASN.1 never cut mid-way
- Target chunk: 200–350 tokens (max 500)
- 8 chunk types: section, event_def, ASN.1, table, figure, annex, history, formula

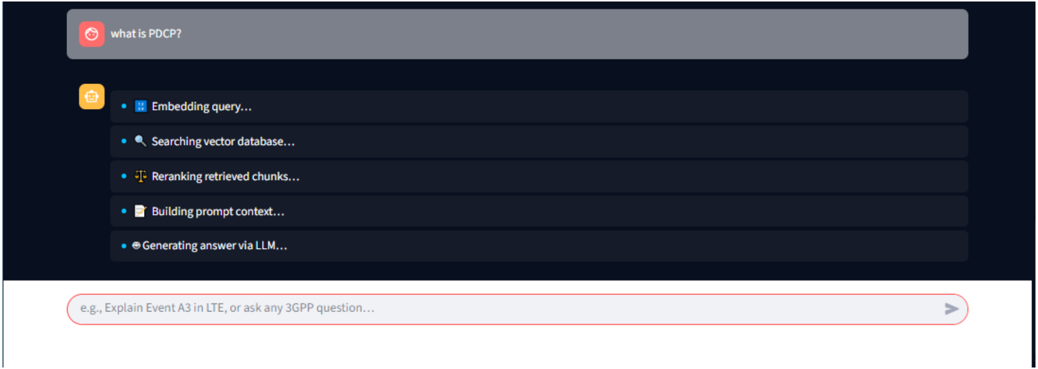
Retrieval & Generation

- **BGE-large-en-v1.5** embedder → local **Qdrant** vector store
- Top-20 retrieved → cross-encoder reranker → top-5 sent to LLM
- **Anti-hallucination prompt**: model must say *"cannot find in specifications"* rather than guess
- Metadata filter: restrict to specific TS / Release

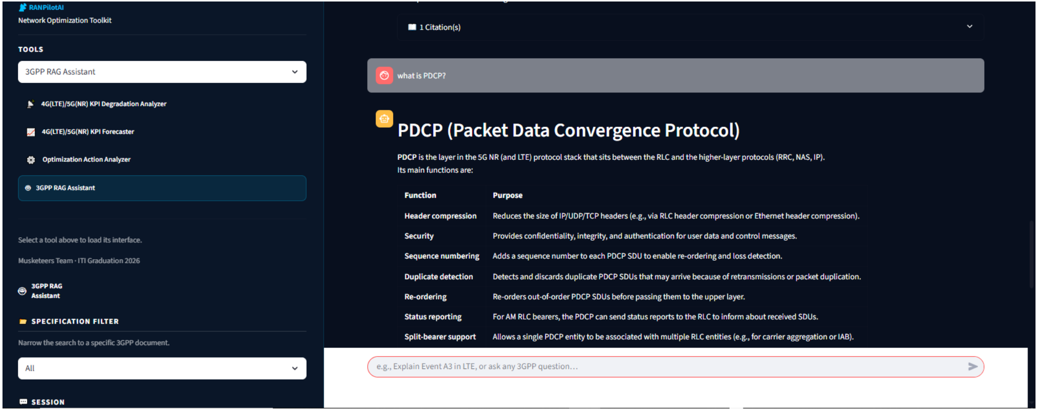
RAG Assistant — Main Interface



Live Query — Pipeline Execution Steps

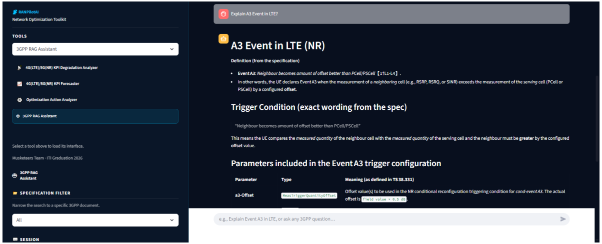


Retrieved Chunks & Source Citations

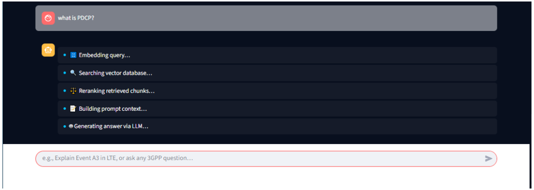


Answer Panel & Backend Response

Citation-Backed Answer Panel



Backend Pipeline Stages



Mohab Tarek

Software Testing ▪ Real-Case Experimental Results ▪ Conclusion ▪ Future Work

Testing Strategy & Test Coverage

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Unit & Integration Testing

[Fill in your content here]

Real-Case Dataset Description

[Fill in your content here]

KPI Degradation — Experimental Results

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`screenshots/mohab/results_degradation.png`

Forecasting — Experimental Results

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`screenshots/mohab/results_forecasting.png`

Action Analyzer — Experimental Results

Configuration Change Event

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`screenshots/mohab/results_action_before.png`

KPI Impact Measured

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`screenshots/mohab/results_action_after.png`

RAG Assistant — Experimental Results

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`screenshots/mohab/results_rag.png`

Conclusion

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Future Work

[Fill in your content here]